

CATEGORICAL NOMINAL VARIABLE IN A MULTIPLE LINEAR REGRESSION

EXAMPLE: $\ln(\text{Profit}) \sim \text{Wealth} + \text{Strategy}$

$\downarrow$ numerical $\downarrow$ categorical with levels A, B, C

THEORETICAL MODEL $\text{Profit} = \beta_0 + \beta_1 \text{Wealth} + \beta_B \mathbf{I}_B + \beta_C \mathbf{I}_C + \varepsilon$

$\downarrow$
 $\mathbf{I}_B$ and $\mathbf{I}_C$ are indicator functions:

$\left\{ \begin{array}{l} \mathbf{I}_B = 1 \text{ if strategy B is} \\ \text{adopted for the customer} \\ \mathbf{I}_B = 0 \text{ otherwise} \end{array} \right.$

$\downarrow$
ESTIMATED MODEL :

If strategy A

$$\hat{\text{Profit}}_i = -11.97 + 0.063 \cdot \text{Wealth}_i$$

If strategy B

$$\hat{\text{Profit}}_i = (-11.97 + 52.81) + 0.063 \text{Wealth}_i$$

If strategy C

$$\hat{\text{Profit}}_i = (-11.97 + 112.28) + 0.063 \text{Wealth}_i$$

The estimated model corresponds to THREE REGRESSION LINES that share the SAME SLOPE but have DIFFERENT INTERCEPT

These are the coefficients that appear in the R studio output.

COEFFICIENT INTERPRETATION :

For the SAME WEALTH LEVEL the EXPECTED PROFIT of a customer for whom STRATEGY B is applied is 52.81 HIGHER than that of a customer for whom STRATEGY A is applied.

For the SAME WEALTH LEVEL the EXPECTED PROFIT of a customer for whom STRATEGY C is applied is 112.28 HIGHER than that of a customer for whom STRATEGY A is applied.

CATEGORICAL EXPLANATORY VARIABLES IN A MULTIPLE LINEAR REGRESSION

EXAMPLE: $\ln(\text{Profit}) \sim \underbrace{\text{Wealth}}_{\text{numerical}} * \underbrace{\text{Strategy}}_{\text{categorical with levels A, B, C}} \leftarrow \text{interaction term}$

THEORETICAL MODEL

$$\text{Profit} = \beta_0 + \beta_1 \text{Wealth} + \beta_B I_B + \beta_C I_C +$$

$$\beta_{\text{Wealth}:B} I_B \text{Wealth} + \beta_{\text{Wealth}:C} I_C \text{Wealth}$$

ESTIMATED MODEL :

If strategy A

$$\hat{\text{Profit}}_i = -20.73 + 0.066 \text{Wealth}_i$$

If strategy B

$$\hat{\text{Profit}}_i = (-20.71 + 69.7) + (0.066 - 0.005) \text{Wealth}_i$$

If strategy C

$$\hat{\text{Profit}}_i = (-20.71 - 0.0278) + (0.066 - 0.0279) \text{Wealth}_i$$

The estimated model corresponds to THREE
REGRESSION LINES with DIFFERENT INTERCEPTS
and DIFFERENT SLOPES

SLOPE COEFFICIENT INTERPRETATION

For every additional unit of WEALTH the EXPECTED PROFIT of a customer increases by:

0.066 if STRATEGY A is adopted

0.060 if STRATEGY B is adopted

0.038 if STRATEGY C is adopted